

Eq. (27.6.11), so that Eq. (27.6.10) may be written

$$N_x = 1 - L. \quad (27.6.12)$$

Thus the coefficient c_λ of X in Eq. (27.6.7) is correctly given by the tree approximation and is therefore what it was in the original Lagrangian, simply $c_\lambda = 1$, while the coefficient L_λ of the X -independent term is given by one-loop diagrams only. Putting this all together, we have

$$\begin{aligned} \mathcal{L}_\lambda^\# = & \left[\mathcal{A}_\lambda(\Phi, \Phi^\dagger, V, X, X^\dagger, Y, Y^\dagger, \mathcal{D} \cdots) \right]_D + 2 \operatorname{Re} \left[Y f(\Phi) \right]_{\mathcal{F}} \\ & + \frac{1}{2} \operatorname{Re} \left[(X + L_\lambda) \sum_{A\alpha\beta} \epsilon_{\alpha\beta} W_{A\alpha L} W_{A\beta L} \right]_{\mathcal{F}}, \end{aligned} \quad (27.6.13)$$

where L_λ is the one-loop contribution. Setting $Y = 1$ and $X = 1/g^2$ then gives Eq. (27.6.2), with $g_\lambda^{-2} = g^{-2} + L_\lambda$. As shown in Section 18.3, the leading order contribution to $\lambda dg_\lambda/d\lambda$ is the same function of g_λ whatever renormalization scheme is used to define this coupling, so to one-loop order we must have

$$\lambda dg_\lambda/d\lambda = b g_\lambda^3, \quad (27.6.14)$$

where b is the same coefficient of g^3 as in the renormalization group equation of Gell-Mann and Low. The solution is Eq. (27.6.3), completing the proof.

In theories with a $U(1)$ gauge superfield V_1 , the Lagrangian may contain a Fayet–Iliopoulos term (27.2.7):

$$\mathcal{L}_{\text{FI}} = \xi \left[V_1 \right]_D. \quad (27.6.15)$$

It is easy to see that the coefficient ξ of such a term is not renormalized.⁷ If the corresponding coefficient ξ_λ in the Wilsonian Lagrangian density did depend on the gauge couplings or the couplings in the superpotential, then when we replace the original Lagrangian (27.6.1) with a Lagrangian (27.6.4) involving the external superfields X and Y , this term in the Wilsonian Lagrangian would be required by supersymmetry to take the form

$$\mathcal{L}_{\text{FI}\lambda}^\# = \left[\xi_\lambda(X, Y, X^*, Y^*) V_1 \right]_D, \quad (27.6.16)$$

with ξ_λ a function with a non-trivial dependence on X and/or Y and/or their adjoints. But such a term would not be gauge-invariant, because, according to Eq. (27.2.18), a gauge transformation shifts V_1 by a chiral superfield $i(\Omega - \Omega^*)/2$, and although the D -term of a chiral superfield vanishes, the product of $i(\Omega - \Omega^*)/2$ and ξ_λ is not chiral for general gauge transformations if ξ_λ has any dependence on other superfields. There actually are diagrams that make contributions to ξ_λ that are independent